ASSIGNMENT-REGRESSION ALGORITHM

1. PROBLEM STATEMENT OR REQUIREMENT:

A client's requirement is, he wants to predict the insurance charges based on the several parameters. The Client has provided the dataset of the same. As a data scientist, you must develop a model which will predict the insurance charges.

2. DATASET:

- The dataset has 6 columns age, sex, bmi, children, smoker and charges.
- We have to predict the charges based on age, sex, bmi, children and smoker.
- Therefore, charges is the target variable or output (Dependent).
- Age, sex, bmi, children and smoker are the features or input (Independent).

3. PRE-PROCESSING METHODS:

- The dataset contains 2 columns with **nominal categorical data** so we have to convert them to numerical data.
- This is done by performing **One-Hot enconding.**

4. MODEL:

- Since the dataset contains multiple inputs we will create model using Multiple linear regression, Support vector machine, Decision tree and Random forest algorithms
- In these models we will select the best model for deployment based on R2_score

5.MODEL EVALUATION:

1. **Multiple Linear Regression**
   R2_score = 0.78

2. **Support Vector Machine**

S. NO	Hyper Parameter (C)	Linear kernel (R2 score)	Ploy kernel(R2 score)	Rbf kernel(R2 score)	Sigmoid kernel(R2 score)
1	1	-0.01	-0.07	-0.09	-0.07
2	0.1	-0.08	-0.08	-0.08	-0.08
3	0.01	-0.08	-0.08	-0.08	-0.08
4	10	0.46	0.038	-0.03	0.03
5	100	0.62	0.61	0.32	0.51
6	1000	0.76	0.85	0.81	0.28

3.Decision Tree

S.NO	CRITERION	SPLITTER	MAX_DEPTH	R2_SCORE
1	Squared error	Best	3	0.87
2	Squared error	Best	5	0.82
3	Squared error	Best	15	0.69
4	Squared error	Random	3	0.77
5	Squared error	Random	5	0.87
6	Squared error	Random	15	0.7
7	Friedman mse	Best	3	0.87
8	Friedman mse	Best	5	0.82
9	Friedman mse	Best	15	0.69
10	Friedman mse	Random	3	0.75
11	Friedman mse	Random	5	0.84
12	Friedman mse	Random	15	0.68
13	Absolute error	Best	3	0.86
14	Absolute error	Best	5	0.85
15	Absolute error	Best	15	0.66
16	Absolute error	Random	3	0.77
17	Absolute error	Random	5	0.78
18	Absolute error	Random	15	0.72
19	Poisson	Best	3	0.87
20	Poisson	Best	5	0.83
21	Poisson	Best	15	0.71
22	Poisson	Random	3	0.73
23	Poisson	Random	5	0.84
24	Poisson	Random	15	0.71

4.Random forest

S.NO	N_ESTIMATORS	CRITERION	MAX_FEATURES	R2 SCORE
1	100	Squared error	sqrt	0.86
2	200	Squared error	sqrt	0.86
3	50	Squared error	sqrt	0.86
4	100	Squared error	log2	0.87
5	200	Squared error	log2	0.87
6	50	Squared error	log2	0.86
7	100	Absolute error	sqrt	0.87
8	200	Absolute error	sqrt	0.87
9	50	Absolute error	sqrt	0.87
10	100	Absolute error	log2	0.87
11	200	Absolute error	log2	0.87
12	50	Absolute error	log2	0.86
13	100	Friedman mse	sqrt	0.87
14	200	Friedman mse	sqrt	0.87

15	50	Friedman mse	sqrt	0.86
16	100	Friedman mse	log2	0.87
17	200	Friedman mse	log2	0.86
18	50	Friedman mse	log2	0.86
19	100	Poisson	sqrt	0.87
20	200	Poisson	sqrt	0.87
21	50	Poisson	sqrt	0.86
22	100	Poisson	log2	0.87
23	200	Poisson	log2	0.87
24	50	Poisson	log2	0.86

5.RESULT:

- After performing a lot of trial and error on the dataset for different Machine Learning model. I have achieved highest accuracy of 87% in various models.
- **As a conclusion I have selected Random Forest model as the best model with the parameters**
    1. Criterion = squared error
    2. N_estimators = 100
    3. Max_features = log2

Reason:

1. **Highest accuracy:** Random forest model achieved highest accuracy of 87%.
2. **Stability:** While smaller forests (10 or 50 trees) also reached similar scores, a forest of **100 trees** was selected to ensure greater model stability.
3. **Computational Efficiency:** The use of `max_features='log2'` ensures that the model remains efficient during inference.